DP-ShiftBench

How Distribution Shift Governs
the Value of Pretraining
for Private Learning

Project Plan

Course: PrivAI 2026 (IIT Madras)

Instructor: Krishna Pillutla

Problem Statement: 1.6 — The Effect of Distribution Shifts
on Pretraining for Private Learning

Project Type: Original empirical research (with theoretical validation)

Target: Workshop paper at NeurIPS 2026
(Privacy in ML / Distribution Shift workshops)

March 2026

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1 Key Dates

Milestone	Date	Status
Project Proposal (2 pages, NeurIPS format)	March 27, 2026	TODO
Mid-Review Report (4 pages)	April 17, 2026	TODO
Project Presentation	April 29 – May 4, 2026	TODO
Final Report (9 pages)	May 11, 2026	TODO

Table 1: Project milestones and deadlines.

2 Paper Thesis

*When a model is pretrained on public data and finetuned with DP on private data, how does the distribution shift between these two datasets affect the privacy-utility tradeoff? We build a controlled benchmark (**DP-ShiftBench**), discover which divergence metrics predict DP degradation, and empirically test whether the interaction between shift and privacy is additive or multiplicative.*

3 Three Core Contributions

3.1 Contribution 1: DP-ShiftBench — A Controlled Benchmark Suite

A reusable, open-source benchmark where distribution shift is a **continuous, tunable knob**.

3.1.1 1A. Image Track: SVHN → MNIST

Exp. ID	Pretrain Data	Pretrain Method	Finetune Data (DP)	Shift Control Knob	Pri.
IMG-00	None (random init)	—	MNIST full	Baseline (no pre-train)	P0
IMG-01	SVHN full	Supervised	MNIST full	Full domain gap	P0
IMG-02	SVHN digits {0–4}	Supervised	MNIST digits {0–4}	Matched class subset	P0
IMG-03	SVHN digits {5–9}	Supervised	MNIST digits {0–4}	Mismatched class subset	P0
IMG-04	SVHN full	Autoencoder	MNIST full	Task gap	P1
IMG-05	SVHN 10%	Supervised	MNIST full	Data scarcity	P1
IMG-06	SVHN 25%	Supervised	MNIST full	Data scarcity	P1
IMG-07	SVHN 50%	Supervised	MNIST full	Data scarcity	P1
IMG-08	CIFAR-10	Supervised	MNIST full	Larger domain gap	P1
IMG-09	SVHN + aug.	Supervised	MNIST full	Style shift	P2
IMG-10	FashionMNIST	Supervised	MNIST full	Same format, diff. semantics	P2

Model: ResNet-18 (or small CNN with ~500K parameters for faster iteration).

DP Settings for each experiment:

- $\varepsilon \in \{0.5,\ 1.0,\ 2.0,\ 4.0,\ 8.0,\ \infty$ (non-private)$\}$
- $\delta = 10^{-5}$
- Clipping norm $C \in \{0.1,\ 1.0\}$ (pick best after initial sweep)
- Batch size: 256 (with gradient accumulation if needed)
- Epochs: 10–20 for finetuning

3.1.2 1B. Language Track: Controlled Markov Chains

Exp. ID	Pretrain Distribution	Finetune Distribution (DP)	Shift Control Knob	Pri.
`LANG-00`	None	Markov(P_1)	Baseline	P0
`LANG-01`	Markov(P_1)	Markov(P_1)	Zero shift	P0
`LANG-02`	Markov(P_1)	Markov($0.9P_1 + 0.1P_u$)	$\alpha = 0.1$	P0
`LANG-03`	Markov(P_1)	Markov($0.7P_1 + 0.3P_u$)	$\alpha = 0.3$	P0
`LANG-04`	Markov(P_1)	Markov($0.5P_1 + 0.5P_u$)	$\alpha = 0.5$	P0
`LANG-05`	Markov(P_1)	Markov($0.3P_1 + 0.7P_u$)	$\alpha = 0.7$	P0
`LANG-06`	Markov(P_1)	Markov(P_2), independent	$\alpha = 1.0$ (max shift)	P0
`LANG-07`	Small Transformer	Small Transformer (DP)	Real distribution shift	P1
`LANG-08`	GPT-2 small	GPT-2 small (DP)	Real domain shift	P2

Markov Chain Details:

- Vocabulary size: 100 tokens
- Order: $k = 3$ (start), then $k = 4$, $k = 5$
- Transition probabilities P_1: randomly generated, then fixed
- Shift control: $P_2(\alpha) = (1 - \alpha)\, P_1 + \alpha\, P_{\text{uniform}}$
- This gives **exact** KL divergence computation (no estimation needed)
- Train a small LSTM or Transformer on sequences from these chains

Model: 2-layer LSTM (for Markov) or GPT-2 small (for real text).
DP Settings: Same ε grid as the image track.

3.1.3 1C. Benchmark Package Design

```
dp_shift_bench/
    __init__.py
    datasets/
        image_shifts.py       # SVHN/MNIST/CIFAR shift generators
        markov_chains.py      # Controllable Markov chain distributions
        text_shifts.py        # Real text domain shift loaders
    models/
        image_models.py       # ResNet-18, SmallCNN
        language_models.py    # LSTM, small Transformer
    training/
        dp_trainer.py         # Unified DP-SGD training loop (Opacus)
        standard_trainer.py   # Non-private pretraining loop
    metrics/
        divergences.py        # All divergence measures
        evaluation.py         # Accuracy, perplexity, etc.
    configs/
        image_experiments.yaml
```

```
        language_experiments.yaml
    runners/
        run_experiment.py     # Single experiment runner
        run_sweep.py          # Full grid sweep
```

3.2 Contribution 2: Divergence-as-Predictor Analysis

For **every** experiment, compute multiple divergence measures between pretrain and finetune data, then analyze which metric best predicts the DP finetuning performance.

3.2.1 Divergence Metrics to Compute

Metric	**Implementation**	**Works For**	**Pri.**
KL Divergence	Exact (Markov), `scipy` est. (neural)	Language (exact), Images (est.)	**P0**
Reverse KL	Same as above	Both	**P0**
Jensen–Shannon Div.	`scipy`	Both	**P0**
Total Variation	Exact (Markov), est. (neural)	Both	**P0**
MAUVE Score	`mauve-text` (Pillutla et al.)	Language	**P0**
FID	`clean-fid`	Images	**P0**
MMD (RBF kernel)	Custom implementation	Both	**P1**
Proxy $\mathcal{A}$-distance	Domain classifier, $2(1-2\epsilon)$	Both	**P1**
Wasserstein Distance	`POT` library	Both	**P1**
CKA	Custom (repr. similarity)	Both	**P2**

3.2.2 Analysis Plan

1. **Correlation analysis:** For each metric, compute Pearson/Spearman correlation with DP test accuracy across all experiments.
2. **Predictive modeling:** Fit accuracy$(\varepsilon, d) = f(\varepsilon, d)$ where d is the divergence.
3. **Rank the metrics:** Which metric has the highest predictive power?
4. **Cross-domain validation:** Does the best predictor on images also work for language?

3.2.3 Key Plots

- **Scatter plot:** Divergence metric (x) vs. DP accuracy (y), colored by ε — one plot per metric
- **Correlation heatmap:** Metrics × Epsilons, showing R^2 values
- **Radar chart:** Comparing predictive power across all metrics

3.3 Contribution 3: Additive vs. Multiplicative Interaction

3.3.1 The Competing Hypotheses

H_{additive} (current theory — Bassily et al. 2023):

$$\text{Error}(\varepsilon, d) = C_1 \cdot \frac{1}{\varepsilon} + C_2 \cdot d + C_3 \tag{1}$$

The DP cost ($1/\varepsilon$) and domain shift cost (d) contribute independently.

$H_{\text{multiplicative}}$ (open question):

$$\text{Error}(\varepsilon, d) = C_1 \cdot \frac{d}{\varepsilon} + C_2 \tag{2}$$

Distribution shift **amplifies** the privacy cost — more shift means you need even more privacy budget.

$H_{\text{threshold}}$ (Setlur et al. 2025 "Goldilocks zone"):

$$\text{Error}(\varepsilon, d) = \begin{cases} C_1/\varepsilon + C_2 \cdot d & \text{if } d < d_{\text{threshold}} \\ C_1/\varepsilon + C_3 & \text{if } d \geq d_{\text{threshold}} \text{ (pretrain useless)} \end{cases} \tag{3}$$

There is a critical shift threshold beyond which public data stops helping.

3.3.2 Experimental Design

Run a **full 2D grid**:

- $\varepsilon \in \{0.5,\ 1.0,\ 2.0,\ 4.0,\ 8.0,\ \infty\}$
- shift_level $\in \{0.0,\ 0.1,\ 0.2,\ 0.3,\ 0.5,\ 0.7,\ 1.0\}$ (for Markov: α parameter)

This gives a $6 \times 7 = 42$ cell grid. Each cell is repeated 3 times for error bars.

Total: 126 runs for language, 126 for images.

3.3.3 Analysis Plan

1. **3D surface plot:** accuracy(ε, shift) — the "hero figure"
2. **Model fitting:** Fit H_{additive}, $H_{\text{multiplicative}}$, $H_{\text{threshold}}$ to the data. Report AIC/BIC.
3. **Phase diagram / heatmap:** 2D plot showing "gain from pretraining" = acc(pretrained)−acc(random_init) as a function of (ε, shift). Identify the Goldilocks zone boundary.
4. **Interaction test:** Two-way ANOVA for significant interaction effect between ε and shift.

4 Bonus Enhancements (If Time Permits)

4.1 Enhancement A: Membership Inference Attacks Under Shift

Motivation: Does distribution shift affect *empirical* privacy leakage?
Method: Run LiRA or basic shadow-model MIA on DP-finetuned models across shift levels.
Key question: Does shift amplify or reduce MIA success rate?
Priority: P2 **Effort:** ~3 days

4.2 Enhancement B: Pretraining Data Selection

Motivation: Given a pool of public data, can we *select* a subset that minimizes DP finetuning loss?
Method: Use divergence metrics to rank public data subsets, pretrain on top-k, evaluate.
Priority: P2 **Effort:** ~2 days

4.3 Enhancement C: LoRA vs. Full Finetuning Under Shift

Motivation: Does parameter-efficient finetuning (LoRA) behave differently under shift?
Method: Repeat key experiments with LoRA finetuning instead of full finetuning.
Priority: P2 **Effort:** ~2 days

4.4 Enhancement D: Theoretical Bound Derivation (Stretch)

Motivation: Derive a simple theoretical bound connecting distribution shift to DP finetuning error for convex/linear models.

Approach:

- For linear models, the DP-SGD error depends on gradient norm / clipping threshold.
- Under distribution shift, the gradient norms change predictably.
- Can derive:

$$\text{excess\_risk}_{\text{DP}}(\text{shift}) = \mathcal{O}\left(\frac{d \cdot C \cdot \sqrt{T} \cdot \sigma}{n}\right) + \mathcal{O}(\text{shift\_distance}) \tag{4}$$

  where the σ term is the DP noise.
- Check if the interaction is additive or multiplicative for this simple case.

Priority: P2 (but very high impact if achieved) **Effort:** ~5 days

5 Technical Stack

Component	Tool	Notes
DP Training	Opacus v1.4+	PyTorch-based DP-SGD
Deep Learning	PyTorch 2.x	Core framework
Image Models	torchvision (ResNet-18, CNN)	Pretrained or from scratch
Language Models	HuggingFace Transformers	GPT-2 for real text
MAUVE Score	`mauve-text`	Pillutla et al.
FID	`clean-fid`	Image quality metric
Optimal Transport	`POT`	Wasserstein distance
Experiment Tracking	Weights & Biases	Logging all runs
Config Management	Hydra + YAML	Reproducible experiments
Compute	V100 (16 GB)	Single GPU sufficient
Plotting	matplotlib + seaborn	Publication-quality figures
Paper	LaTeX (NeurIPS template)	As required by course

Table 5: Technical stack overview.

6 Project Structure

```
PrivInAI-Project-2026/
|
|-- PROJECT_PLAN.md                # This document
|-- README.md                      # Repo overview
|
|-- dp_shift_bench/                # Main benchmark package
|   |-- __init__.py
|   |-- datasets/
|   |   |-- image_shifts.py        # SVHN/MNIST/CIFAR loaders
|   |   |-- markov_chains.py       # Controllable Markov chains
|   |   |-- text_shifts.py         # Real text domain shift
|   |-- models/
|   |   |-- image_models.py        # ResNet-18, SmallCNN
```

```
|    |    |-- language_models.py   # LSTM, small Transformer
|    |-- training/
|    |    |-- dp_trainer.py        # DP-SGD loop (Opacus)
|    |    |-- standard_trainer.py  # Non-private pretraining
|    |    |-- utils.py             # LR schedulers, etc.
|    |-- metrics/
|    |    |-- divergences.py       # All divergence measures
|    |    |-- evaluation.py        # Accuracy, perplexity
|    |-- configs/
|    |    |-- default.yaml
|    |    |-- image_experiments.yaml
|    |    |-- language_experiments.yaml
|    |-- runners/
|         |-- run_single.py        # Run one experiment
|         |-- run_sweep.py         # Full experiment grid
|         |-- run_divergences.py   # Compute divergence metrics
|
|-- experiments/                   # Experiment scripts
|-- analysis/                      # Analysis notebooks
|-- results/                       # Saved results (gitignored)
|-- paper/                         # LaTeX source for reports
|-- tests/                         # Unit tests
|-- requirements.txt
|-- setup.py
```

7 Execution Timeline

Aggressive — targeting April 17 mid-review with near-complete results.

7.1 Week 1: March 22–28 (Proposal Week)

Deadline: Proposal due March 27

Day	Task	Parallelizable?
Mar 22–23	Set up repo, project structure, dependencies	No
Mar 22–23	Read core papers: Li et al., Ganesh et al.	No
Mar 23–24	Implement `markov_chains.py` — data generator	Yes (Agent 1)
Mar 23–24	Implement `image_shifts.py` — SVHN/MNIST loaders	Yes (Agent 2)
Mar 23–24	Implement `divergences.py` — KL, JSD, TV	Yes (Agent 3)
Mar 24–25	Implement `dp_trainer.py` — Opacus training loop	No (core)
Mar 25–26	Run first baselines (IMG-00, IMG-01, LANG-00, LANG-01)	Yes
Mar 26–27	**Write 2-page proposal**	No
Mar 27	**SUBMIT PROPOSAL**	—

7.2 Week 2: March 29 – April 4 (Core Experiments)

Day	Task	Parallel?	Est. Time
Mar 29–30	Run full image grid (IMG-00 to IMG-08)	Yes (Agent 1)	∼8h GPU

Day	Task	Parallel?	Est. Time
Mar 29–30	Run full Markov language grid (LANG-00 to LANG-06)	Yes (Agent 2)	~2h GPU
Mar 29–30	Implement remaining divergences (MAUVE, FID, MMD, Wasserstein)	Yes (Agent 3)	—
Mar 31 – Apr 1	Compute ALL divergence metrics for ALL dataset pairs	Yes	~4h
Apr 1–2	Run 2D grid ($\varepsilon\times$ shift) — language	Yes (Agent 1)	~6h
Apr 1–2	Run 2D grid ($\varepsilon\times$ shift) — images	Yes (Agent 2)	~12h
Apr 2–3	Implement proxy $\mathcal{H}$-divergence (domain classifier)	Yes (Agent 3)	—
Apr 3–4	Run P1 experiments (IMG-04–07, LANG-07)	Yes	~8h

7.3 Week 3: April 5–11 (Analysis & Extra Experiments)

Day	Task	Parallel?
Apr 5–6	Contribution 2 analysis: correlation, predictive modeling	Yes (Agent 1)
Apr 5–6	Contribution 3 analysis: model fitting, ANOVA	Yes (Agent 2)
Apr 5–6	Generate ALL hero figures	Yes (Agent 3)
Apr 7–8	Enhancement A (MIA under shift) — if time permits	Optional
Apr 7–8	Enhancement C (LoRA comparison) — if time permits	Optional
Apr 9–10	Run failed/missing experiments, collect final results	—
Apr 10–11	Create results summary tables	—

7.4 Week 4: April 12–17 (Mid-Review Report)

Day	Task
Apr 12–13	Write mid-review Sections 1–2 (intro, literature survey with math)
Apr 13–14	Write mid-review Section 3 (experimental setup, benchmark design)
Apr 14–15	Write mid-review Section 4 (results so far)
Apr 15–16	Create all figures and tables for mid-review
Apr 16–17	Polish, proofread, finalize
Apr 17	**SUBMIT MID-REVIEW**

7.5 Weeks 5–6: April 18 – May 4 (Presentation + Remaining Work)

Task	Timeline
Enhancement experiments (MIA, LoRA, data selection)	Apr 18–22
Package `dp_shift_bench` for release	Apr 22–25
Prepare presentation slides	Apr 25–28
PRESENT	Apr 29 – May 4

7.6 Week 7: May 5–11 (Final Report)

Task	Timeline
Write full 9-page report	May 5–9
Final figures, proofreading	May 9–10
SUBMIT FINAL REPORT	May 11

8 Key References

8.1 Core References (from course document)

Ref	Authors	Title	Year	Relevance
[21]	Li, Tramer, Liang, Hashimoto	Large language models can be strong differentially private learners	2022	Why pretraining helps DP
[22]	Ganesh, Haghifam, Nasr, Oh, Steinke, Thakkar, Thakurta, Wang	Why is public pretraining necessary for private model training?	2023	Two-phase hypothesis
[10]	Pillutla, Swayamdipta, Zellers, et al.	MAUVE: Measuring the Gap Between Neural Text and Human Text	2021	Distribution gap measurement
[11]	Pillutla, Liu, Thickstun, et al.	MAUVE Scores for Generative Models: Theory and Practice	2023	Extended MAUVE theory
[23]	Netzer et al.	Reading digits in natural images with unsupervised feature learning (SVHN)	2011	SVHN dataset
[24]	Sakarvadia et al.	Mitigating memorization in language models	2025	Language memorization

8.2 Additional Key References (from literature search)

Authors	Title	Year	Relevance
Setlur, Thaker, Ullman	Lower Bounds for Public-Private Learning under Distribution Shift	2025	Goldilocks zone theory
Bassily, Cortes, Mao, Mohri	Differentially Private Domain Adaptation with Theoretical Guarantees	2023	Additive bound (we test this)
Ben-David, Blitzer, Crammer, et al.	A theory of learning from different domains	2010	Domain adaptation theory
Auddy, Cai, Chakraborty	Minimax and Adaptive Transfer Learning under Distributed DP	2025	Minimax rates for private transfer
Dong, Roth, Su	Gaussian Differential Privacy	2022	GDP framework
Tobaben et al.	On the Efficacy of DP Few-shot Image Classification	2023	DP with pretrained features
De, Berrada, Hayes, Smith, Balle	Unlocking High-Accuracy DP Image Classification through Scale	2022	Scaling + pretraining for DP
Li, Guo, Li, Fan, Hu, Liu	PrivLM-Bench	2024	Existing DP benchmark (no shift)
Yichuan, Kotevska, Reshniak	Assessing MIA under Distribution Shifts	2024	MIA + shift

9 Risk Register

Risk	Likelihood	Impact	Mitigation
GPU time insufficient for full grid	Low	High	Prioritize P0 experiments; reduce grid
Markov chain results too "toy"	Medium	Medium	Include real text experiments (LANG-07/08)
No clear winner among divergence metrics	Medium	Low	Valid negative result
Additive vs. multiplicative inconclusive	Medium	Medium	Report fitted models; let data speak
Opacus bugs / compatibility issues	Low	High	Pin versions; test early
Proposal too ambitious for reviewers	Low	Low	Clearly mark P0/P1/P2 priorities

Table 14: Risk register with mitigation strategies.

10 Success Criteria

10.1 Minimum Viable Paper (Course Project)

- ☐ DP-ShiftBench with image track (P0 experiments) working
- ☐ DP-ShiftBench with language Markov track (P0 experiments) working
- ☐ At least 4 divergence metrics computed and correlated with DP accuracy
- ☐ 2D grid ($\varepsilon\times$ shift) for either images or language
- ☐ Phase diagram / hero figure
- ☐ Clear narrative about which metric predicts DP degradation

10.2 Full Paper (Workshop Submission)

All of the above, **plus**:

- ☐ Both image AND language tracks complete
- ☐ All 6+ divergence metrics computed and compared
- ☐ Formal additive vs. multiplicative hypothesis test
- ☐ Cross-domain validation (best predictor on images also works for language?)
- ☐ Open-source benchmark package released
- ☐ At least one enhancement (MIA, LoRA, or data selection)

10.3 Stretch Goals (Full Conference Paper)

- ☐ Theoretical bound derivation for linear models
- ☐ Enhancement D complete
- ☐ Extended to more realistic datasets (CIFAR-100, real NLP)
- ☐ Temporal shift experiments